Universidad Nacional Abierta y a Distancia
Vicerrectoría Académica y de Investigación
Course: Advanced Machine Learning
Code: 203238427

Learning Guide - Phase 3 – Ensemble learning

1. Information of the Phase 3

Table 1. *Description.*

Aspect	Description
1. Type of activity	Independent
2. Evaluation moment	**Intermediate**
3. Managing unit	Escuela de Ciencias Básicas Tecnología e Ingeniería ECBTI
4. Score of the Choose an option	**100**
5. The activity starts on:	Wednesday, March 18, 2026
6. The activity ends on:	Tuesday, April 14, 2026
7. Student independent work hours	**20**

2. Detailed Description of the Learning Activity

Through this activity, the following learning outcome is expected to be achieved*:*

Learning Outcome 3: Apply ensemble models, both generative and non-generative, to improve predictive performance in machine learning tasks.

The activity consists of:

The following activity is an individual assignment where the student must use the same dataset applied in ***Phase 2 - Practical component - Simulated practice*** and propose the most suitable ensemble methods. You must submit a single document with the 4 exercises developed in the format **Appendix 1 – Template_Phase3**.

The following materials and resources are required for the development of this activity:

- Desktop or laptop computer. Internet connection.
- Python software installed on the computer or Google Colab account.
- John Hearty. (2016). Advanced Machine Learning with Python. Packt Publishing. (pp 201-221) https://research-ebsco-com.bibliotecavirtual.unad.edu.co/plink/b08cc392-af99-3021-bba2-cdd015b6e5b0
- Lior Rokach. (2010). Pattern Classification Using Ensemble Methods (Vol. 00075). World Scientific.(pp 19-40) https://research-ebsco-com.bibliotecavirtual.unad.edu.co/plink/603e1fdc-22e0-38b5-9cef-a14c0681cfbd
- Du, K.-L., Zhang, R., Jiang, B., Zeng, J., & Lu, J. (2025). Foundations and Innovations in Data Fusion and Ensemble Learning for Effective Consensus . Mathematics, 13(4), 587. (pp 1 - 49) https://doi.org/10.3390/math13040587
- Härner, S. (2022). Comparing ensemble methods (stacking and voting) using Naïve Bayes and decision trees in diabetes detection. [Bachelor's thesis, Halmstad University]. DiVA portal. (pp 14 – 19) https://www.diva-portal.org/smash/get/diva2:1701964/FULLTEXT01.pdf
- Appendix 1 – Template_Phase3.ipynb

Exercise 1: Voting and Stacking Methods

You can also use the codes presented on the following GitHub:

• https://github.com/PacktPublishing/Hands-On-Ensemble-Learning-with-Python/tree/master/Chapter03

• https://github.com/PacktPublishing/Hands-On-Ensemble-Learning-with-Python/tree/master/Chapter04

The student must implement in a Python Notebook the Bagging Method, with the data of Phase 2, according to the chosen item in Phase 2, the methods of Voting (Hard and Soft) with the classification dataset and the method of Stacking with the regression dataset. Calculate the metrics (Mean Squared Error, Mean Absolute Error, R2 for regression and accuracy, precision, recall, F1-score for classification) and learning

curves for these methods. In a Markdown cell presents the analysis of these methods indicating which is more effective.

Exercise 2: Bagging Method

The student must implement in a Python Notebook the Bagging Method, with the data of Phase 2, according to the chosen item in Phase 2, generating 10 bootstrap samples. Calculate the metrics for these methods (Mean Squared Error, Mean Absolute Error, R2 for regression and accuracy, precision, recall, F1-score for classification) as well as the learning curve, and in a Markdown, cell present the analysis of your results.

Exercise 3: Boosting Method

The student must implement in a Python Notebook the Boosting Methods (AdaBoost, Gradient Boosting and XGBoost), with the data set of Phase 2, according to the chosen item in Phase 2.
Calculate the metrics for these methods (Mean Squared Error, Mean Absolute Error, R2 for regression and accuracy, precision, recall, F1-score for classification) as well as the learning curve and in a Markdown, cell present the analysis of your results.

Exercise 4: Random Forest

The student must implement in a Python Notebook the Random Forest Method, with the data set of exercise 2 of Phase 2, according to the chosen item in Phase 2. Calculate the metrics for these methods (Mean Squared Error, Mean Absolute Error, R2 for regression and accuracy, precision, recall, F1-score for classification) as well as the learning curve and in a Markdown cell present the analysis of your results.

3. Guidelines for Developing and Submitting Learning Evidence

Learning evidence refers to the actions, products, or observable processes that are done or delivered to demonstrate acquired capabilities, skills, aptitudes, and attitudes. These serve to allow the teacher to assess and evaluate student performance effectively.

The evidence to be developed individually are:

- The development of the four exercises, which you will have to perform in documents
- with ipynb extension, as requested in the delivery conditions.

For developing and submitting evidence, please consider the following guidelines:

1. All group members must participate with their contributions in the development of the activity.

2. Each group should designate one member to be responsible for submitting the requested product in the environment specified by the teacher.

3. Before submitting the requested product, the group should check that it complies with all the requirements indicated in this activity guide.

4. Only group members who have actively contributed during the allocated time for the activity should be included as authors of the requested product.

Please note that all individual or group written products must comply with "spelling and mechanics" standards and the submission conditions defined in this activity guide.

Regarding the use of references, consider that the product of this activity must comply with the standards of Choose an option.

In any case, comply with the referencing standards and avoid academic plagiarism. To do so, you can support yourself by reviewing your written products using the Turnitin tool available in the virtual campus.

4. Academic Situations

Consider that in Agreement 029 of December 13, 2013, Article 99, the following actions are considered as offenses against academic order, among others: item e) "Plagiarism, that is, presenting as one's own the entirety or part of a work, paper, document, or invention created by another person. It also includes the use of false citations or references, or proposing citations where there is no match between the citation and the reference," and item f) "Reproducing or copying, for profit, educational materials or results of research products that have intellectual property rights reserved for the University."

The academic sanctions that the student will face are as follows:

a) In cases of proven academic fraud in the respective academic work or evaluation, the grade imposed will be zero points without prejudice to the corresponding disciplinary sanction.

b) In cases related to proven plagiarism in academic work of any nature, the grade imposed will be zero points, without prejudice to the corresponding disciplinary sanction.